

CYBERSECURITY ASSESSMENT REPORT

Phishing Email Detection & Awareness System

Task 02 | Future Interns — Cyber Security Internship

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Task	Task 02 — Phishing Email Detection
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1. Executive Summary

Phishing attacks remain one of the most dangerous cybersecurity threats facing individuals and organizations worldwide. Attackers use spoofed emails, fake websites, malicious attachments, and social engineering tactics to steal sensitive information such as passwords, financial credentials, and confidential business data.

This report presents a practical analysis of five email samples — both phishing and legitimate — collected for educational purposes. Each email was examined for phishing indicators, sender authenticity, suspicious links, and psychological manipulation tactics.

The goal of this project was to simulate a real-world phishing analysis workflow and build practical skills in cybersecurity investigation and awareness reporting.

2. Introduction

Phishing is a form of social engineering where attackers impersonate trusted organizations — banks, tech companies, employers — to trick users into giving up sensitive information or taking harmful actions.

Common phishing tactics include:

- Fake account suspension or verification warnings
- Urgent payment or wire transfer requests
- Malicious attachments disguised as HR or payroll documents
- Fake login pages that steal credentials
- Business Email Compromise (BEC) impersonating executives

This project analyzes five email samples and explains clearly how to spot the difference between a real email and a phishing attempt.

3. Project Objectives

- Identify phishing indicators in real-looking email samples
- Analyze spoofed sender domains and fake URLs
- Understand how attackers use urgency and fear to manipulate users
- Classify emails by risk level (Safe, Suspicious, Phishing, Critical)
- Create clear cybersecurity awareness documentation
- Recommend practical prevention strategies

4. Tools & Technologies Used

Tool	Purpose
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MXToolbox	Email header analysis and DNS verification
Google Admin Toolbox	Advanced email header investigation
Browser Developer Tools	Inspect suspicious URLs and redirects
GitHub	Repository hosting and project documentation
Canva	Create project visuals and awareness screenshots
Kali Linux	Cybersecurity analysis environment

5. Project Methodology

The following steps were followed during this analysis:

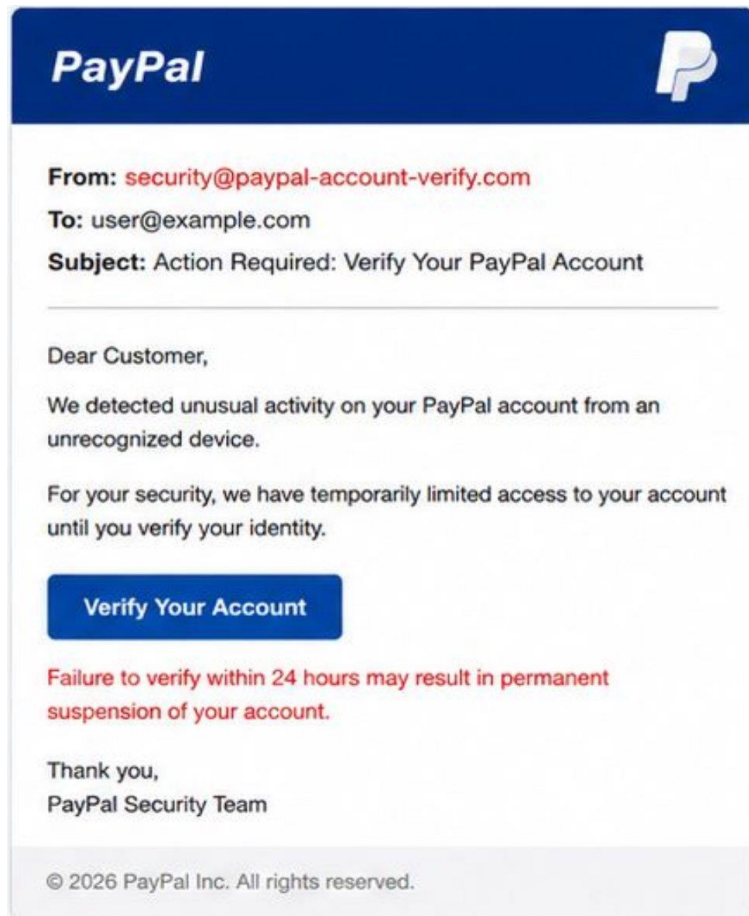
1. Collected publicly available phishing email samples for educational analysis.
2. Identified phishing indicators such as spoofed domains, suspicious links, urgent language, and malicious attachments.
3. Performed email header and sender analysis using online investigation tools.
4. Inspected suspicious URLs and phishing domains.
5. Classified email samples into Safe, Suspicious, Phishing, and Critical categories.
6. Prepared awareness documentation and prevention guidelines based on findings.

6. Phishing Email Sample Analysis

Five email samples were analyzed in this project. Each sample is presented with its screenshot, classification, risk level, indicators, and a plain-English explanation of what makes it safe or dangerous.

Sample A — PayPal Account Verification Scam

Classification: PHISHING	Risk Level: HIGH
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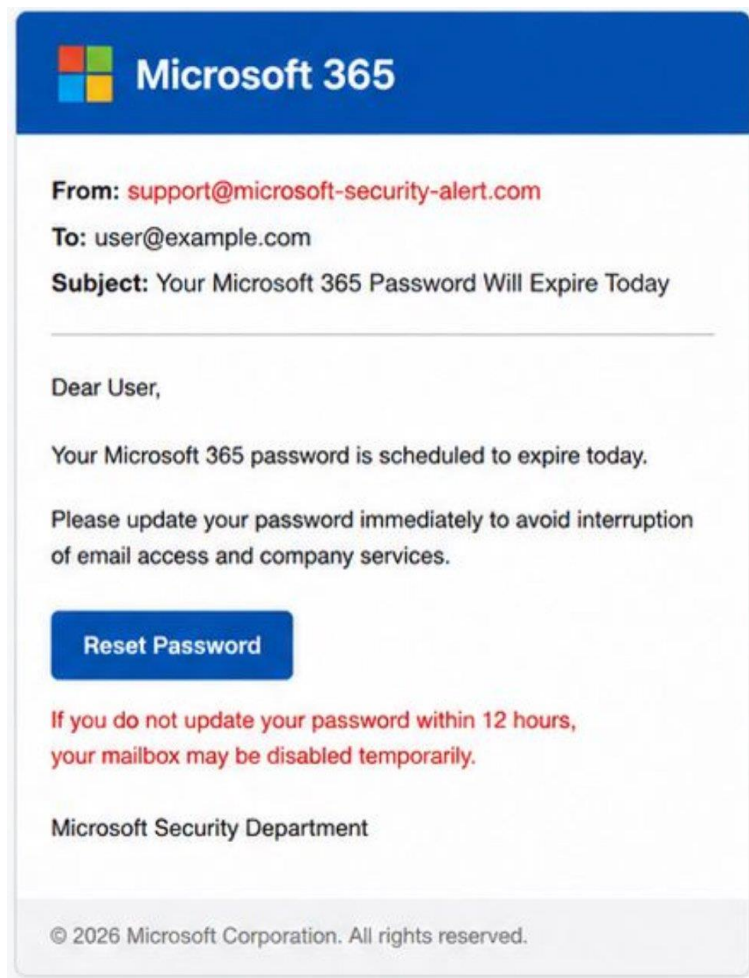
**Indicators Identified:**

- Sender domain is 'paypal-account-verify.com' — not the real paypal.com
- Creates fear by claiming unusual account activity
- Uses a 'Verify Your Account' button that links to a fake login page
- Threatens permanent account suspension within 24 hours
- Generic greeting: 'Dear Customer' instead of your real name
- Designed to steal your PayPal username and password

Analysis:

This email pretends to be from PayPal's security team. The sender domain is fake — the real PayPal only sends emails from @paypal.com. The email uses urgency (24-hour deadline) and fear (account suspension) to push users into clicking the fake 'Verify Your Account' button, which leads to a credential harvesting page.

Sample B — Microsoft 365 Password Reset Scam

Classification: PHISHING**Risk Level: HIGH****Indicators Identified:**

- Sender domain is 'microsoft-security-alert.com' — not microsoft.com
- Claims password will expire today to create panic
- Fake 'Reset Password' button leads to a malicious website
- Warns mailbox will be disabled in 12 hours to force action
- Generic greeting: 'Dear User' instead of your actual name
- Attempts to steal Microsoft 365 login credentials

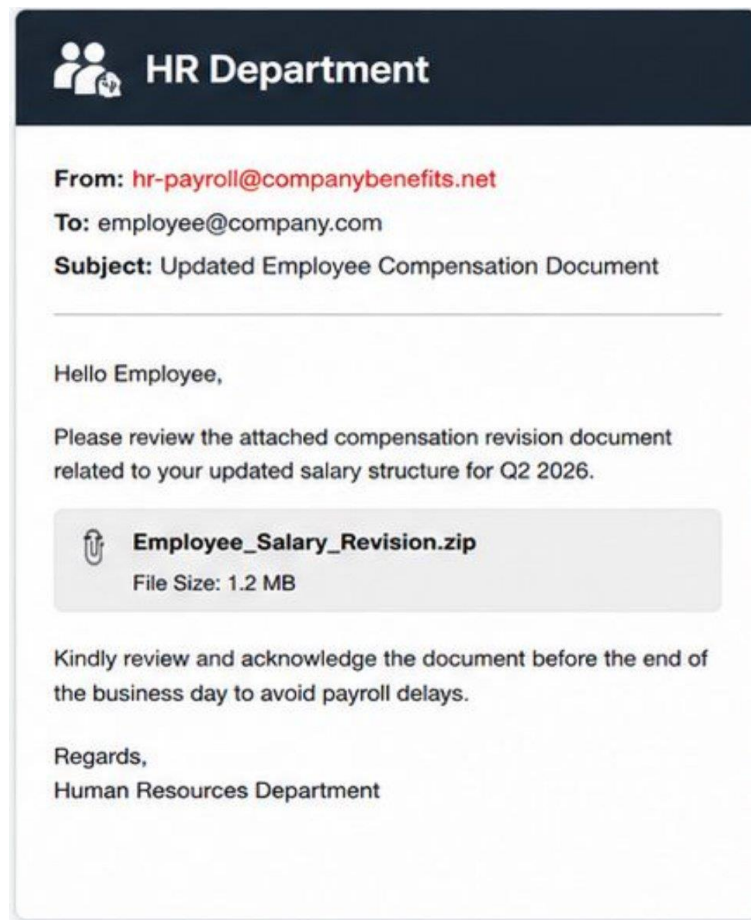
Analysis:

This email mimics a Microsoft 365 security alert. The domain 'microsoft-security-alert.com' is not owned by Microsoft — it is a fake domain registered by attackers. The combination of a tight 12-hour deadline and threat of mailbox shutdown is a classic social engineering technique to rush users into clicking without thinking.

Sample C — HR Salary Document (Suspicious Attachment)

Classification: SUSPICIOUS

Risk Level: MEDIUM



Indicators Identified:

- Sender domain is 'companybenefits.net' — not an official company domain
- Attaches a .zip file named 'Employee_Salary_Revision.zip' — could contain malware
- Generic greeting: 'Hello Employee' instead of your name
- Creates urgency around payroll to pressure quick action
- Legitimate HR departments rarely send salary docs as .zip attachments
- Could be used to deliver ransomware or keyloggers

Analysis:

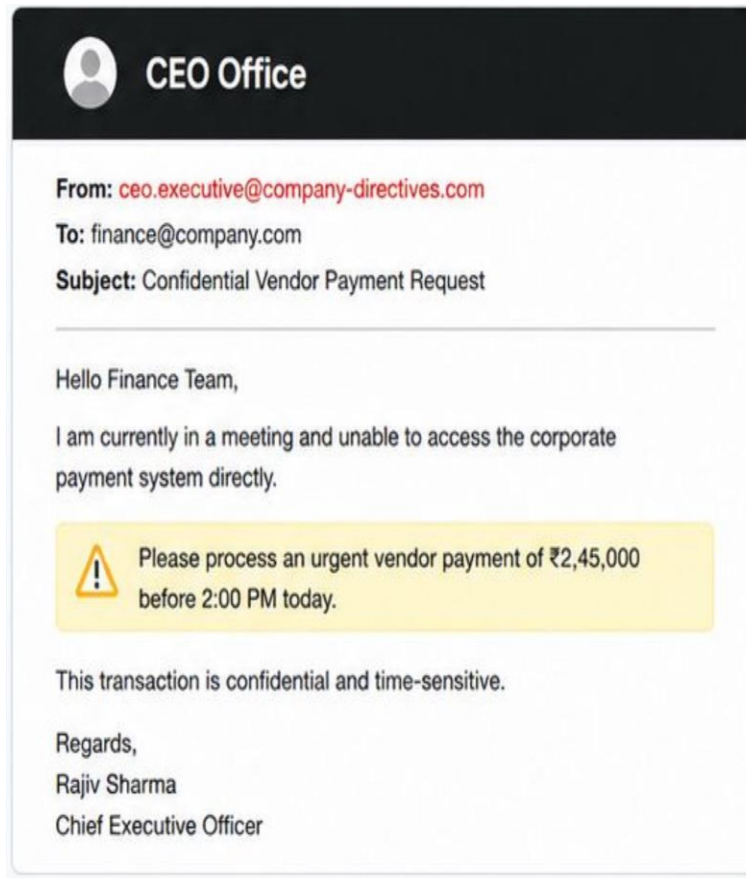
This email appears to come from the HR Department but uses a suspicious external domain. The biggest red flag is the .zip attachment — legitimate HR compensation documents are

typically sent as PDF files through official company portals, not as compressed archives from external email addresses. Opening this attachment could install malware on your device.

Sample D — Business Email Compromise (CEO Fraud)

Classification: PHISHING

Risk Level: CRITICAL



Indicators Identified:

- Sender domain is 'company-directives.com' — not the real company domain
- Impersonates the CEO (Rajiv Sharma) to gain trust
- Requests an urgent payment of ₹2,45,000 before 2:00 PM
- Claims to be in a meeting and unable to access payment system directly
- Marks the transaction as 'confidential' to prevent verification
- Classic Business Email Compromise (BEC) financial fraud attack

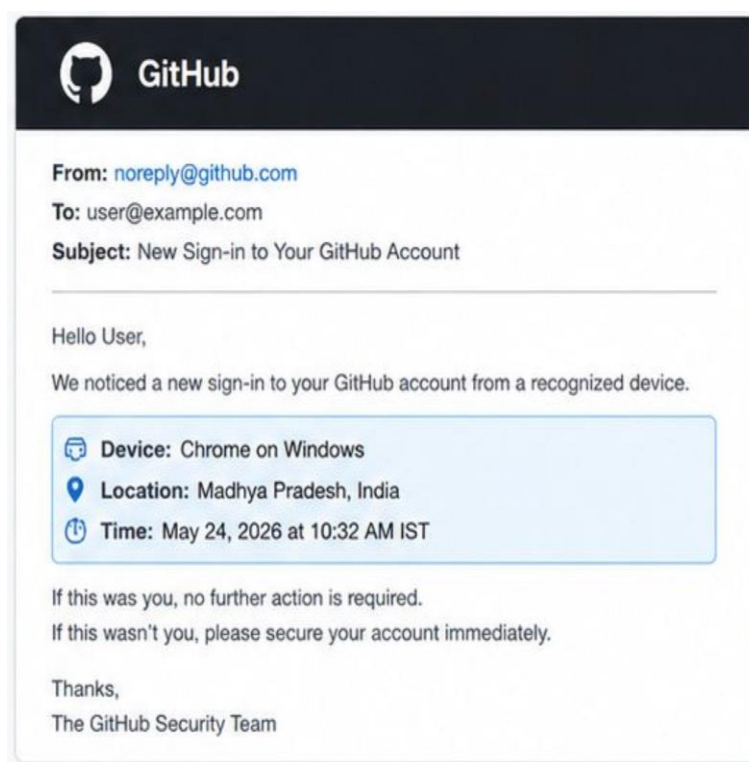
Analysis:

This is one of the most dangerous types of phishing — Business Email Compromise (BEC). The attacker pretends to be the company's CEO and puts time pressure on the finance team to process a wire transfer without following normal approval procedures. The 'confidential' label is used to discourage the victim from verifying the request with colleagues. Always verify unusual financial requests through a separate phone call to the actual executive.

Sample E — GitHub Security Notification (Legitimate)

Classification: SAFE

Risk Level: LOW



Indicators Identified:

- Sender domain is 'github.com' — the real, official GitHub domain
- Professional and clear formatting with no urgency tactics
- Shows specific sign-in details: device, location, and exact timestamp
- Provides a calm, helpful message: no threats or alarming language
- No suspicious buttons or links asking for credentials
- Legitimate security notification style from a verified sender

Analysis:

This email is a genuine security notification from GitHub. The key difference from phishing emails is that it uses the real github.com domain, provides specific contextual details about the sign-in event, and does not pressure the user. It simply informs the user about a login and tells them what to do if it was not them — no threats, no fake urgency, no suspicious links.

7. Email Header Analysis

Email headers contain technical information about where an email really came from. Most email clients hide this by default, but it can reveal spoofing attempts.

Key indicators found during header analysis of the phishing samples:

- SPF (Sender Policy Framework) failures — the sending server was not authorized to send on behalf of the claimed domain
- Missing DKIM (DomainKeys Identified Mail) signatures — legitimate emails are digitally signed
- Suspicious relay servers — emails routed through unusual third-party servers
- Spoofed Return-Path addresses — replies would go to a different address than the displayed sender
- Sender domain mismatches — the 'From' name and actual email domain do not match

Simple rule: If the 'From' name says 'Microsoft' but the actual email is 'support@microsoft-security-alert.com', it is a phishing email.

8. URL & Domain Analysis

Phishing emails rely on fake domains that look similar to real ones. Here are examples from the analyzed samples:

Phishing Domain Used	Real Domain	Tactic
security@paypal-account-verify.com	service@paypal.com	Extra words added to look official
support@microsoft-security-alert.com	support@microsoft.com	Fake subdomain with 'security' keyword
hr-payroll@companybenefits.net	hr@yourcompany.com	External domain pretending to be internal
ceo.executive@company-directives.com	ceo@yourcompany.com	Fake CEO domain with corporate-sounding name

9. Risk Classification Summary

Email Sample	Risk Level	Classification
PayPal Verification Scam	HIGH	Phishing

Microsoft Password Reset	HIGH	Phishing
BEC — CEO Fraud	CRITICAL	Phishing (Critical)
HR Salary Attachment	MEDIUM	Suspicious
GitHub Security Notification	LOW	Safe

10. Common Phishing Indicators

The following warning signs were found repeatedly across the analyzed phishing emails:

Indicator	What to Look For
Spoofed Sender	The 'From' name looks real, but the actual email domain is fake or misspelled
Urgency & Fear	Phrases like 'Act now', 'Within 24 hours', 'Your account will be suspended'
Fake Buttons/Links	Buttons that say 'Verify' or 'Reset' but link to malicious websites
Generic Greetings	'Dear User' or 'Dear Customer' instead of your actual name
Suspicious Attachments	ZIP files, EXE files, or unexpected documents from unknown senders
Confidentiality Pressure	Telling you not to discuss the request with others — a BEC red flag
Too Good or Too Scary	Either a scary threat or an unexpected reward to push you into action

11. Awareness & Prevention Guidelines

Following these simple practices can significantly reduce your risk of falling victim to a phishing attack:

- Always check the sender's actual email domain, not just the display name
- Never click links in suspicious emails — go directly to the website by typing the URL
- Never download unexpected attachments, especially .zip, .exe, or .rar files
- Enable Multi-Factor Authentication (MFA) on all important accounts
- When in doubt about a financial request, verify by calling the person directly
- Report suspicious emails to your IT or security team immediately
- Keep your browser, email client, and operating system updated
- Attend regular cybersecurity awareness training
- Use anti-phishing email filters and security software
- Remember: Legitimate companies will NEVER ask for your password by email

12. Conclusion

This project provided hands-on experience in analyzing phishing emails, identifying attack techniques, and understanding how cybercriminals exploit human psychology through urgency, fear, and impersonation.

The five email samples demonstrated a wide range of phishing tactics — from fake PayPal and Microsoft alerts to dangerous Business Email Compromise attacks targeting financial departments. The one legitimate email (GitHub notification) showed how real service emails are clearly different from phishing attempts.

Key takeaways from this project:

- Always verify the sender's actual email domain before trusting any email
- Urgency and fear are the attacker's most powerful tools — slow down and think
- Technical controls (MFA, email filtering) combined with user awareness are the best defense
- BEC attacks are the most financially dangerous form of phishing and require strict verification procedures

This project also improved practical cybersecurity skills including threat analysis, risk assessment, investigation techniques, and professional security documentation.

13. Disclaimer

All phishing email samples used in this project were created or sourced as publicly available educational examples. They are intended strictly for cybersecurity awareness, analysis, and learning purposes only. No malicious activity or unauthorized testing was performed during this project.